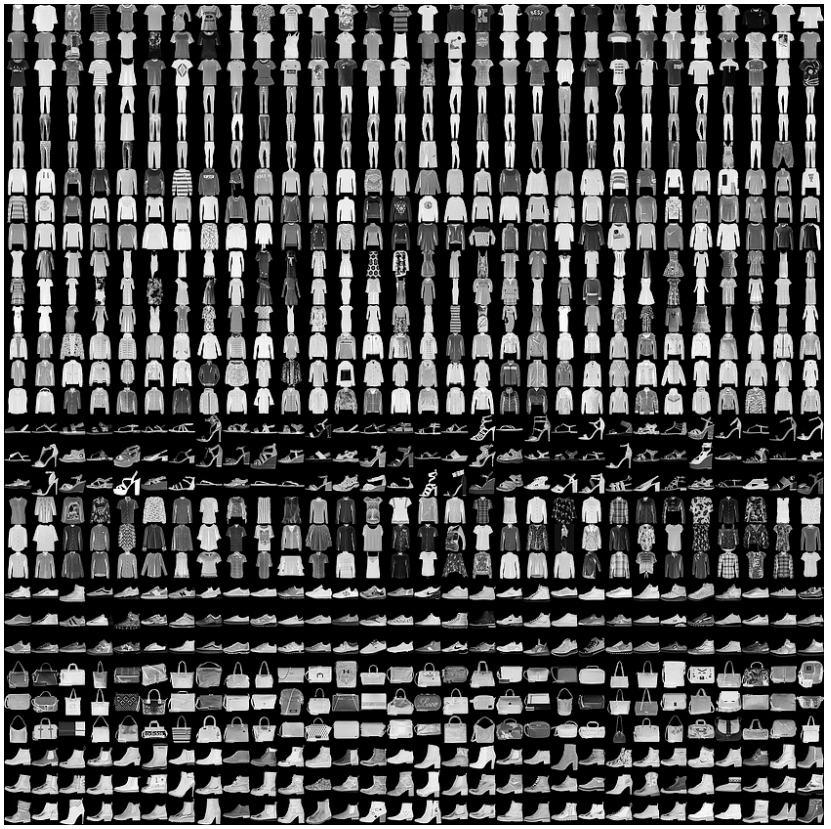
Image Classification

Motivating application: Fashion MNIST

The fashion-mnist dataset consists of 70,000 images of clothing items across 10 categories.

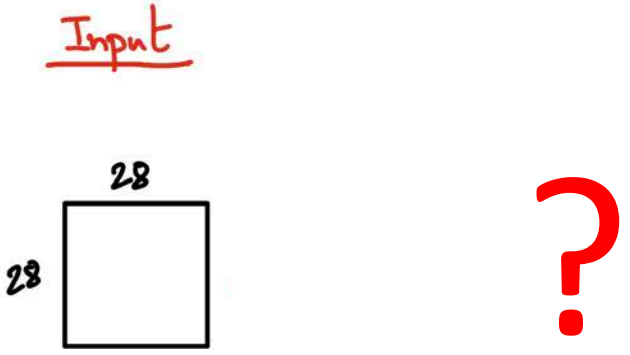
We will build a deep learning network from scratch to classify clothing into these 10 categories with over 90% accuracy!



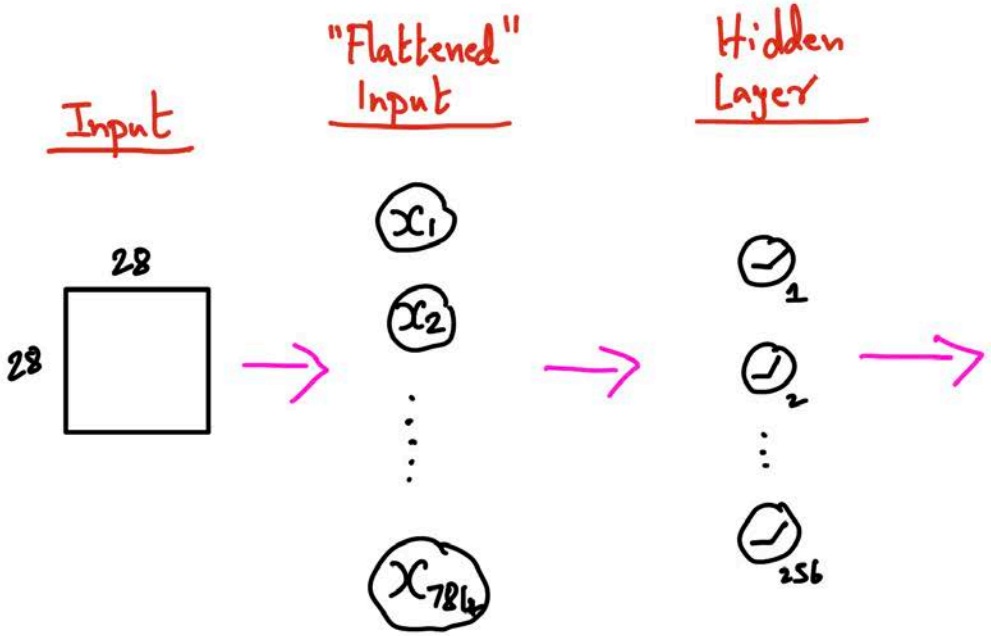
Sample of the Fashion MNIST images by Yuzamei.
Source: Wikimedia Commons. License: CC BY-SA.

Source: https://www.kaggle.com/datasets/zalando-research/fashionmnist

A simple NN to classify grayscale clothing images



A simple NN to classify grayscale clothing images

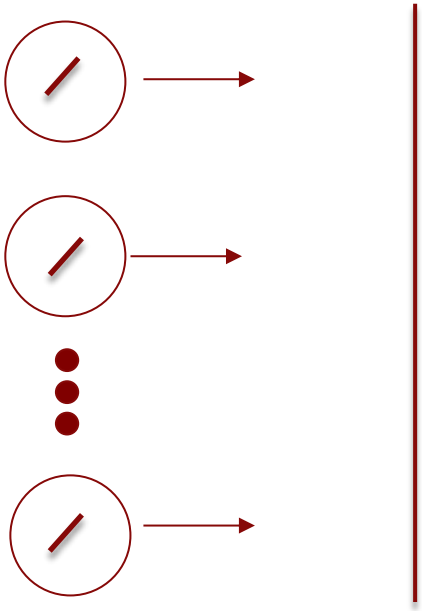


How do we represent a 10-way output?

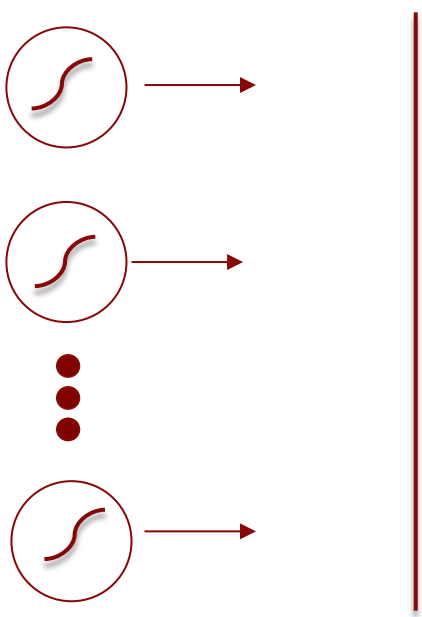
Multi-Class Classification

Suppose the output variable is categorical with 10 levels

We know how to output 10 numbers



We know how to output 10 probabilities



How do we output 10 probabilities that sum to 1.0?

Summary: Output Layers for Regression and Classification

Output Variable	Output Layer
Single number (regression with a single output)	
Single probability (binary classification)	
Vector of *n* numbers (regression with multiple outputs)	Stack of 
Vector of n probabilities that add up to 1 (multi-class classification)	Softmax